



SOMAIYA
VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute permanently affiliated to University of Mumbai.

Academic Year 2025-26(Odd Sem)
Department Of Computer Engineering
Mini Project Log Book
Course Code-PR31

Course Name-Project Based Learning: Mini Project Lab-I

Roll No	Name of Student	Signature of Student
02	Atharva Naik	
14	Vikas Patel	
48	Arin Sutariya	
51	Anshuman Tiwari	

Name of Project- Snap & Report – Civic Complaint Management System

Name of Guide - Mrs. Kavita Bathe

PBL Outcomes

At the end of the course, the student will be able to:

Sr. No	PBL Outcomes
1	Identify problems based on societal /research needs.
2	Apply Knowledge and skill to solve societal problems in a group
3	Develop interpersonal skills to work as member of a group or leader.
4	Analyze the impact of solutions in societal and environmental context for sustainable development
5	Excel in written and oral communication
6	Demonstrate capabilities of self-learning in a group, which leads to lifelong learning
7	Demonstrate project management principles during project work

Continuous Assessment

Distribution of Term work marks for both semesters shall be as below:	Practical Marks
Marks awarded by guide/supervisor based on implementation	10
Peer assessment by team members	5
Marks awarded by review committee	5
Quality of Project report	5

Guidelines for Mini Project

Sr.No	Guide Lines
1	Project based learning Mini Project Lab-1 should be implemented using Java programming (CEXS33)
2	Students shall form a group of 2 to 3 students, while forming a group shall not be allowed less than two or more than three students, as it is a group activity.
3	Students should do survey and identify needs, which shall be converted into problem statement for mini project in consultation with faculty supervisor/internal committee of faculties.
4	Students shall submit implementation plan in the form of Gantt/PERT/CPM chart, which will cover weekly activity of mini project.
5	A logbook to be prepared by each group, wherein group can record weekly work progress, guide/supervisor can verify and record notes/comments.
6	Faculty supervisor may give inputs to students during mini project activity; however, focus shall be on self-learning.
7	Students in a group shall understand problem effectively, propose multiple solution and select best possible solution in consultation with guide/ supervisor
8	Students shall convert the best solution into working model using Java programming.
9	The solution to be validated with proper justification and report to be compiled in standard format of the college.

10	With the focus on the self-learning, innovation, addressing societal problems and entrepreneurship quality development within the students through the Mini Projects, it is preferable that a single project of appropriate level and quality to be carried out in two semesters by all the groups of the students. i.e., Mini Project 1 in semester III and IV.
11	However, based on the individual students or group capability, with the mentor's recommendations, if the proposed Mini Project adhering to the qualitative aspects mentioned above gets completed in odd semester, then that group can be allowed to work on the extension of the Mini Project with suitable improvements/modifications or a completely new project idea in even semester. This policy can be adopted on case-by-case basis

Assessment Index Sheet

Week No	Detailed Explanation	Guide Signature
Week 1	Project Ideation & Topic Selection We began by forming our project group and brainstorming ideas for a mini project that could address real-world problems. After evaluating several ideas, we shortlisted three possible topics and presented them to the evaluation panel. The panel approved “ Snap & Report – Civic Complaint Management System ” as our final topic.	
Week 2	User Survey & Problem Analysis We researched existing civic complaint platforms such as <i>MyGov</i> and <i>Swachh Bharat App</i> to identify key issues. To gather user feedback, we conducted a Google Form survey that received 30 responses. Analysis revealed that users faced delays in issue resolution and preferred mobile-first platforms with image uploads and live tracking.	
Week 3	Requirement Gathering & System Planning Based on survey insights, we documented the system’s functional and non-functional requirements. We defined user roles (Citizen, Officer, Admin), prepared workflow diagrams, and created use-case and data-flow diagrams. Initial UI wireframes were also drafted for the citizen and admin interfaces.	
Week 4	Technology Stack & Architecture Design We analyzed various technologies and finalized the following stack: React + Tailwind CSS for frontend, Django REST Framework for backend, and PostgreSQL + PostGIS for database. We created the system architecture diagram and ER model and received approval from the guide.	

Week No	Detailed Explanation	Guide Signature
Week 5	Frontend Development (Begin) We initialized the React environment and started designing the Citizen App layout, focusing on a responsive, mobile-first design. Login, registration, and navigation components were developed. The admin dashboard structure was also outlined.	
Week 6	Frontend Development (Completion) We developed the full complaint submission form, implemented validation, and added reusable UI components like buttons, cards, and tables. The admin dashboard was updated to include complaint lists and basic analytics placeholders. Internal testing was done to ensure proper responsiveness.	
Week 7	Backend Setup & Database Configuration We set up the Django backend and configured PostgreSQL with PostGIS for location handling. User authentication was implemented using JWT. API endpoints for registration, login, and complaint creation were created and tested using Postman.	
Week 8	Mid-Project Presentation & Feedback We presented the current progress of the project to the college evaluation panel. The demonstration included the working frontend (citizen and admin sides) and partially functional backend. The panel appreciated the UI design and suggested enhancements such as better color contrast, role-based access, and adding notifications for complaint updates. Based on this feedback, we planned necessary modifications for the next phase.	

Week No	Detailed Explanation	Guide Signature
Week 9	Complaint Management Module We developed the complaint model and completed CRUD APIs, including image uploads, category filters, and status updates. Admin users can now view, update, and assign complaints to departments. All APIs were tested successfully.	
Week 10	Notifications & Admin Functions We integrated email notifications for complaint updates and added admin features such as filtering, sorting, and performance tracking. Basic analytics APIs were implemented to display department-wise complaint data on the dashboard.	
Week 11	Integration & Testing The React frontend was integrated with Django APIs using Axios. The complete complaint flow (submission → tracking → resolution) was tested thoroughly. Bugs were fixed, and overall performance and responsiveness were improved for smoother user experience.	
Week 12	Final Refinement & Presentation We added final improvements based on guide feedback, including priority color badges for complaints and better dashboard visualization. The system was deployed locally using Docker and NGINX. During the final presentation, the evaluation panel appreciated the concept, functionality, and UI design.	